

Continuum multiband exact diagonalization of twisted MoTe₂

An ED benchmark for CDW and fractional Chern-insulator neural states

Local numerical benchmark for the Physics-for-AI/QNS programme

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Abstract

We establish a symmetry-resolved exact-diagonalization (ED) benchmark and a continuous-coordinate neural-Bloch experiment for twisted MoTe₂. The continuum calculation reproduces the one-five-band reference energies, while one-band 27-cell ED distinguishes the $\nu = 1/3$ charge-density-wave tower from the approximately three-fold Γ manifold at $\nu = 2/3$. A tracked three-state calculation gives Chern numbers near zero and two, but complete all-sector spectral isolation, large-gauge sewing and twist-grid convergence remain required before assigning a quantized Hall response.

The variational baseline is one generalized determinant of neural orbitals in the full plane-wave continuum basis and follows the Luo–Fu Appendix-B construction. Four 4,128-walker, 120-update runs compare full momentum mixing with an exact- Γ no- M construction. Fresh-walker validation selects no M at both fillings, giving $E/N_e = -36.78317(1019)$ and $-52.39973(1673)$ meV, respectively 0.6101 and 0.3258 meV per particle above five-band ED. An internally equivariant C3-QNS extension at $\nu = 1/3$ was stopped after 116 accepted updates when its training curve plateaued. Although its training energy reaches -37.21882 meV per particle and its independent C_3 -character residual is below 7.2×10^{-5} , precise validation gives only $-36.71907(1850)$ meV, 0.06410 meV above no M and 0.67416 meV above five-band ED. The training-validation shift identifies persistent-walker sampling lag. An unrestricted no- M network with complete-wavefunction P_m projection was then optimized independently in all three C_3 sectors. Fresh-chain validation selects $m = 0$ and gives $E/N_e = -37.194949(10528)$ meV, 0.41178 meV below no M but still 0.198284(10528) meV above five-band ED. At $\nu = 2/3$, the corresponding three-sector scan also selects $m = 0$; precise validation gives $-52.445466(13171)$ meV, 0.04574 meV below no M but 0.280073(13171) meV above five-band ED. The low-cost-to-precise shift is retained as a sampling-sensitivity diagnostic. Full-ratio five-band-projected 1-RDM and direct two-body structure-factor diagnostics are reported, but no present ansatz establishes a variational advantage or a converged natural-orbital/topological result.

Part I

Exact-diagonalization benchmark

1 Motivation

Fractional Chern insulators (FCIs) in moiré semiconductors require simultaneous control of strong Coulomb correlations, band mixing and nonuniform Bloch-state quantum geometry. A quantum neural state (QNS) can reach larger systems and avoid a rigid few-band truncation, but a lower variational energy is not sufficient evidence of the correct phase. The reference calculation must also reproduce symmetry quantum numbers, the torus ground-state counting, momentum occupation and density correlations.

The present benchmark therefore separates two questions. First, the 3×3 multiband calculation tests the continuum Hamiltonian and the variational convergence from one to five retained bands, directly against Fig. 2 of Ref. [1]. Second, the one-band 27-site calculation repeats the phase diagnostics of Supplementary Fig. 5: a Γ, K, K' CDW tower at $\nu = 1/3$ versus an approximately three-fold Γ manifold at $\nu = 2/3$ [1]. The latter momentum counting is supplemented below by flux spectral flow and a gauge-invariant many-body Chern number.

2 Continuum model and quantum geometry

For a fixed spin/valley sector, the single-hole Hamiltonian is

$$H_0 = \begin{pmatrix} \frac{\hbar^2(-i\nabla - \kappa_+)^2}{2m^*} + V_1(\mathbf{r}) & t(\mathbf{r}) \\ t^*(\mathbf{r}) & \frac{\hbar^2(-i\nabla - \kappa_-)^2}{2m^*} + V_2(\mathbf{r}) \end{pmatrix}. \quad (1)$$

The lowest moiré harmonics and plane-wave convention follow Refs. [5, 1]. We use

$$(V, w, \phi, m^*) = (11.2 \text{ meV}, 13.3 \text{ meV}, -91^\circ, 0.62m_e), \quad (2)$$

monolayer lattice constant $a_0 = 0.352$ nm, and eight radial reciprocal shells. The resulting Bloch Hamiltonian

has 61 plane waves per layer. Figure 1 compares the first ten moiré bands at $\theta = 2.6^\circ$ and 3.0° .

For a nondegenerate band n , the quantum geometric tensor is evaluated from velocity matrix elements,

$$Q_{\alpha\beta}^{(n)}(\mathbf{k}) = \sum_{m \neq n} \frac{\langle n | v_\alpha | m \rangle \langle m | v_\beta | n \rangle}{[\varepsilon_n(\mathbf{k}) - \varepsilon_m(\mathbf{k})]^2}, \quad (3)$$

with $g_{\alpha\beta} = \text{Re } Q_{\alpha\beta}$ and $\Omega = -2 \text{Im } Q_{xy}$. Bands 1–3 are calculated on a 49-point linear grid in each reciprocal direction, restricted to the hexagonal Brillouin zone, and smoothly triangulated only for visualization. The shared colorbars retain physical one-dimensional numerical ticks. Values outside the 1.5–98.5 percentile interval are clipped in the rendering, not in the saved data.

3 Projected interaction and ED symmetries

The periodically continued Coulomb interaction is projected into N_b Bloch bands,

$$H = \sum_{n\mathbf{k}} \varepsilon_{n\mathbf{k}} c_{n\mathbf{k}}^\dagger c_{n\mathbf{k}} + \sum_{ab,cd} V_{cd,ab} c_{c\mathbf{k}}^\dagger c_{d\mathbf{k}}^\dagger c_{b\mathbf{k}} c_{a\mathbf{k}}, \quad (4)$$

where $V(\mathbf{q}) = 2\pi e^2/(\epsilon|\mathbf{q}|)$ and the uniform $\mathbf{q} = 0$ component is removed by the neutralizing background. Eight radial reciprocal shells are retained in the form-factor sum.

Charge- $U(1)$ is enforced by fixed hole number, while spin- $U(1)$ is enforced by restricting to the fully polarized sector. Translation symmetry first decomposes the Hilbert space into many-body momentum blocks $\mathcal{H}_K$. At generic momentum, a rotation maps $\mathcal{H}_K$ to the distinct block $\mathcal{H}_{C_n K}$, so point-group symmetry is used only to copy spectra within a momentum orbit. A further C_3/C_6 character decomposition is available only when $C_n K = K + G$, i.e. when the rotation belongs to the little group of K .

The phase-diagnostic cluster has real-space supercell matrix

$$S_{27} = \begin{pmatrix} 3 & -3 \\ 3 & 6 \end{pmatrix}, \quad \det S_{27} = 27, \quad (5)$$

and is therefore the $3\sqrt{3} \times 3\sqrt{3}$ C6 torus rather than a rectangular 3×9 mesh. Its momentum group is written as $\mathbb{Z}_3 \times \mathbb{Z}_9$, with plot index

$$I_k = k_1 + N_1 k_2, \quad N_1 = 3, \quad I_k = 0, \dots, 26. \quad (6)$$

In this convention Γ, K', K occur at $I_k = 0, 9, 18$. The one-band Hilbert space contains $\binom{27}{9} = 4,686,825$ states at $\nu = 1/3$. At $\nu = 2/3$, an exactly verified particle–hole transformation maps 18 particles to 9 holes without changing the spectrum.

4 Multiband benchmark on the 3×3 cluster

Figure 2 summarizes the 3×3 calculation at $\theta = 3^\circ$, $\epsilon = 5$. The ground-state energy decreases monotonically as the retained band space grows. The values in Table 1 reproduce the available $\nu = 2/3$ points in Fig. 2 of Ref. [1] within 0.036 meV per hole. Remote-band mixing is visibly stronger at $\nu = 2/3$.

Table 1: Ground-state energy per hole in meV at $\theta = 3^\circ$, $\epsilon = 5$ on the 3×3 cluster.

N_b	$\nu = 1/3$	$\nu = 2/3$	paper Fig. 2
1	−36.3296	−47.1561	−47.15
2	−37.1158	−50.0774	−50.06
3	−37.3518	−51.8605	−51.83
4	−37.3878	−52.5379	—
5	−37.3932	−52.7255	−52.69

The middle column of Fig. 2 shows the lowest energy in every momentum sector for one-, three- and five-band ED. Energies are plotted as $E_K - E_{\text{GS}}$ in meV, without division by particle number. The right column reports the additive band-resolved weight

$$w_b = \frac{\langle N_b \rangle}{N_h}, \quad \sum_{b=1}^{N_b} w_b = 1. \quad (7)$$

For five bands, the weights are

$$\nu = 1/3 : (0.96469, 0.02953, 0.00523, 0.00049, 0.00005), \quad (8)$$

$$\nu = 2/3 : (0.88952, 0.07591, 0.02535, 0.00763, 0.00159). \quad (9)$$

These are normalized band populations, not overlaps between separately optimized truncated Hilbert spaces.

5 Reproduction of Supplementary Fig. 5

Figure 3 repeats the one-band 27-site benchmark at $\theta = 2.6^\circ$, $\epsilon = 5$, and adds the ground-manifold averaged $n(\mathbf{k})$. At $\nu = 1/3$, the three global low states occur at Γ, K, K' with

$$(E_\Gamma, E_K, E_{K'}) - E_{\text{GS}} = (0, 0.573, 0.573) \text{ meV}. \quad (10)$$

The next state is 1.476 meV above the top of this manifold. The K, K' states fold to Γ in the Brillouin zone of the tripled CDW unit cell, matching the interpretation of Ref. [1].

At $\nu = 2/3$, the three lowest states all lie in the Γ sector,

$$(E_{\Gamma,0}, E_{\Gamma,1}, E_{\Gamma,2}) - E_{\text{GS}} = (0, 0.108, 0.298) \text{ meV}. \quad (11)$$

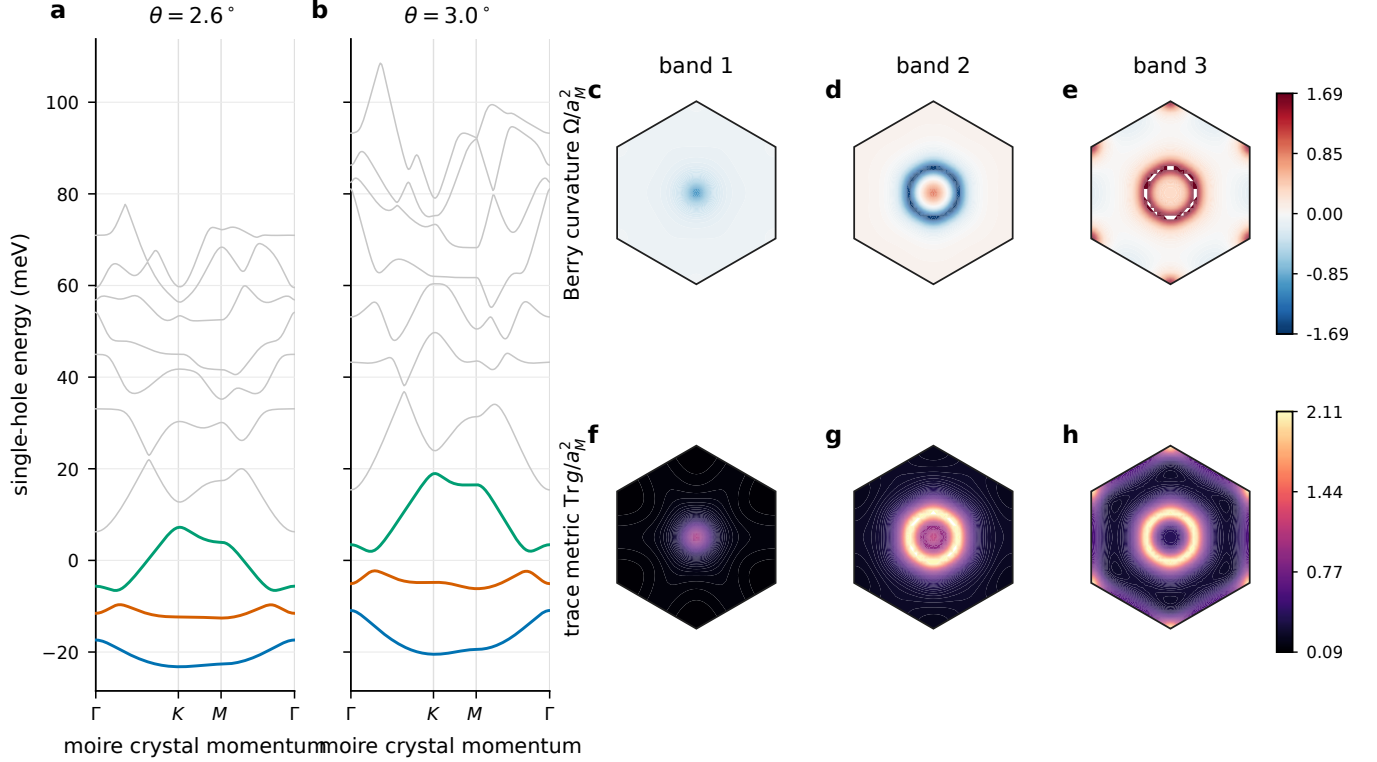


Figure 1: Continuum bands and quantum geometry. (a,b) The first ten moiré bands along Γ - K - M - Γ at $\theta = 2.6^\circ$ and 3.0° ; bands 1–3 are colored blue, orange and green, respectively. (c–e) Berry curvature and (f–h) trace quantum metric for bands 1–3 at $\theta = 2.6^\circ$ across the hexagonal Brillouin zone. Both rows use a common physical color scale with explicit numeric ticks.

Thus the finite-size threefold splitting is small but explicitly nonzero. The gap between the highest manifold state and the next global level is 3.482 meV. This reproduces the torus counting required for the $\nu = 2/3$ FCI/FQAH state.

The absolute energy origins used to construct the two spectra in Fig. 3 are listed explicitly in Table 2. They are the global minima over all 27 translation sectors; the plotted ordinate is obtained by subtracting the corresponding total energy in the fourth column, without dividing the excitation energy by particle number.

Table 2: Absolute one-band, 27-cell ground-state energies used as the zero of the Fig. 3 spectra.

ν	N_h	three-state manifold	E_{GS} (meV)	E_{GS}/N_h (meV)
1/3	9	Γ, K, K'	-428.359163	-47.595463
2/3	18	$\Gamma_0, \Gamma_1, \Gamma_2$	-1034.203144	-57.455730

For each filling, $n(\mathbf{k})$ and the projected static structure factor are averaged over the three-state ground manifold,

$$n(\mathbf{k}) = \langle c_{\mathbf{k}}^\dagger c_{\mathbf{k}} \rangle, \quad (12)$$

$$S(\mathbf{q}) = \frac{1}{N_h} \langle \rho(-\mathbf{q}) \rho(\mathbf{q}) \rangle, \quad \mathbf{q} \neq 0. \quad (13)$$

The connected value $S(0)$ is set to zero. The $n(\mathbf{k})$ sums are exactly 9 and 18. For Fig. 3, the discrete $n(\mathbf{k})$ markers are

placed over an interpolated, weak band-1 Berry-curvature background. The $S(\mathbf{q})$ markers in both rows are placed over the same interpolated band-1 trace-condition deviation $\text{Tr } g_1(\mathbf{q}) - |\Omega_1(\mathbf{q})|$. The background palettes match Fig. 1, while the observable markers use brighter independent color scales.

At $\nu = 1/3$, $S(K) = S(K') = 1.057$, producing the Bragg peaks characteristic of CDW order. At $\nu = 2/3$, the maximum at the same momenta is only 0.197 and is comparable to the surrounding finite- q response; no CDW Bragg peak is present. Together with the three Γ states, this is consistent with the FQAH liquid signature reported in Supplementary Fig. 5.

6 Twisted-boundary spectra and many-body Berry curvature

Let $\mathbf{a}_1, \mathbf{a}_2$ be the primitive moiré real-space vectors and let $\mathbf{b}_a \cdot \mathbf{a}_b = 2\pi\delta_{ab}$. The C_6 -compatible 27-cell ED torus is generated by the columns of

$$S = \begin{pmatrix} 3 & -3 \\ 3 & 6 \end{pmatrix}, \quad |\det S| = 27, \quad (14)$$

$$\mathbf{L}_1 = 3\mathbf{a}_1 + 3\mathbf{a}_2,$$

$$\mathbf{L}_2 = -3\mathbf{a}_1 + 6\mathbf{a}_2.$$

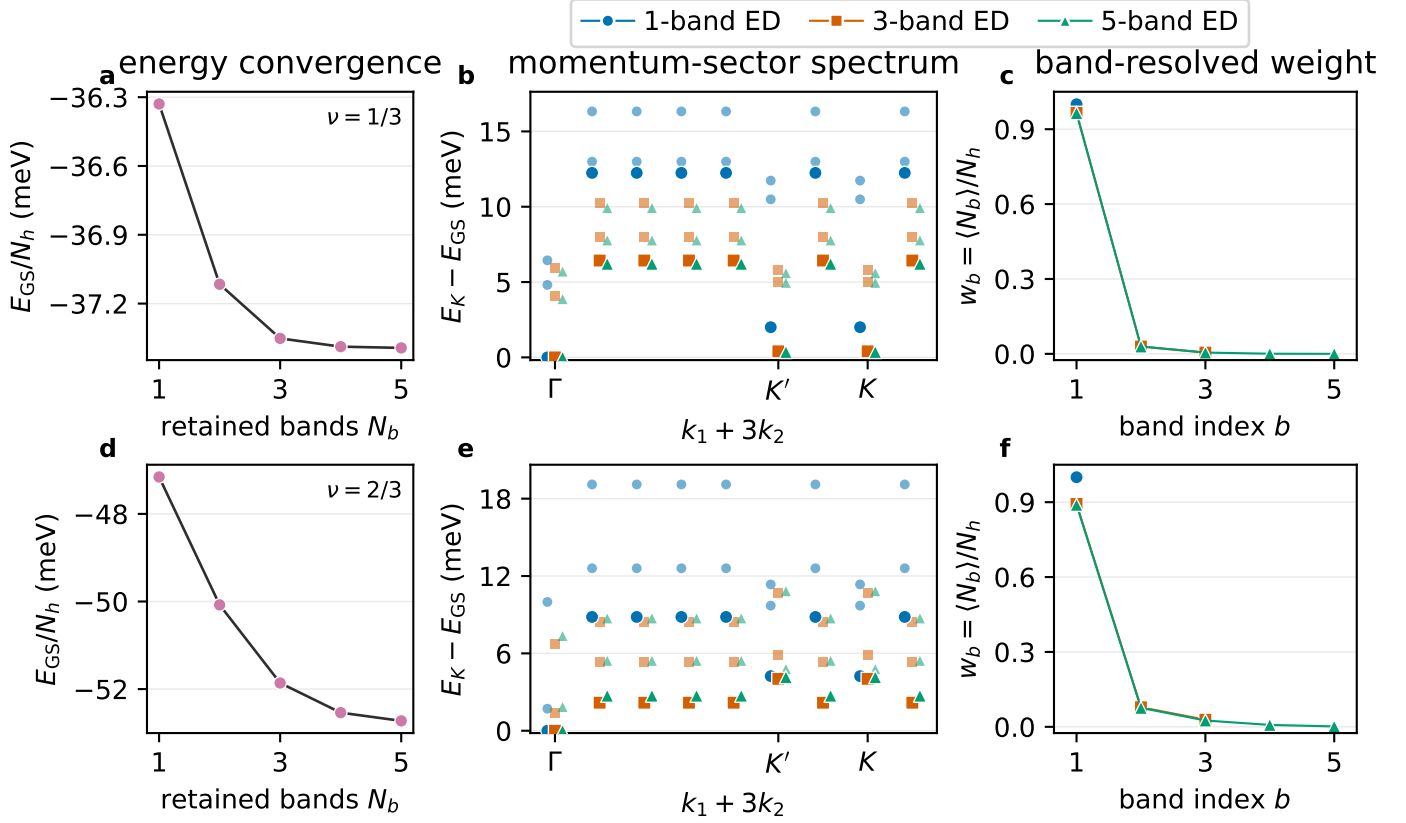


Figure 2: Multiband ED on the 3×3 cluster. Top row: $\nu = 1/3$; bottom row: $\nu = 2/3$. (a,d) Ground-state energy per hole versus retained bands; filling labels are placed in the upper-right blank region. (b,e) The lowest three levels in every momentum sector for one-, three- and five-band ED, plotted as $E - E_{\text{GS}}$ in meV. Larger opaque markers denote sector ground states; the next two levels are smaller and semitransparent. (c,f) Normalized band-resolved ground-state weights. Charge- $U(1)$, spin- $U(1)$ and translations are exact; C_3/C_6 equivalence accelerates the largest five-band sectors.

For every particle i , the many-body twisted boundary condition is

$$\Psi(\dots, \mathbf{r}_i + \mathbf{L}_a, \dots) = e^{i\phi_a} \Psi(\dots, \mathbf{r}_i, \dots), \quad \phi_a \in [0, 2\pi]. \quad (15)$$

Writing a one-particle momentum as $\mathbf{k} = B\mathbf{f}$, with $B = (\mathbf{b}_1, \mathbf{b}_2)$, Eq. (15) gives $S^T \mathbf{f} = \mathbf{m} + \phi/(2\pi)$. Hence the 27 allowed momenta are

$$\mathbf{f}_{\mathbf{m}}(\phi) = S^{-T} \left(\mathbf{m} + \frac{\phi}{2\pi} \right), \quad \mathbf{m} \in \mathbb{Z}^2 / S^T \mathbb{Z}^2, \quad (16)$$

$$\mathbf{k}_{\mathbf{m}}(\phi) = \mathbf{k}_{\mathbf{m}}(0) + \delta \mathbf{k}, \quad \delta \mathbf{k} = BS^{-T} \frac{\phi}{2\pi}. \quad (17)$$

The code enumerates this quotient group with exact integer residues, so precisely 27 inequivalent momenta are retained and every two-body momentum addition is performed modulo $S^T \mathbb{Z}^2$. The torus is therefore implemented periodically without rounding Cartesian momenta.

At every (ϕ_1, ϕ_2) , the continuum Hamiltonian is solved anew at all $\mathbf{k}_{\mathbf{m}}(\phi)$, after which the one-band Coulomb form factors and projected interaction are rebuilt from the shifted Bloch spinors. A projected scattering term is retained only when $\mathbf{k}_1 + \mathbf{k}_2 = \mathbf{k}_3 + \mathbf{k}_4$ in the finite translation group (equivalently up to moiré and supercell reciprocal vectors). Charge- $U(1)$ and translations remain

exact. A generic twist breaks C_3/C_6 , so rotational orbit copying is not used away from little-group-invariant twists. At $\nu = 2/3$, the 18-electron Hamiltonian is evaluated exactly through its nine-hole particle-hole transform, including its constant and induced one-body terms; this leaves the boundary condition and spectrum unchanged.

Because the three lowest states are quasi-degenerate, an individual-eigenvector Berry phase is gauge dependent. We instead use the determinant link of the full three-state subspace,

$$M_{\mu}^{ab}(\phi) = \langle \Psi_a(\phi) | \Psi_b(\phi + \Delta_{\mu}) \rangle, \quad (18)$$

$$U_{\mu}(\phi) = \frac{\det M_{\mu}(\phi)}{|\det M_{\mu}(\phi)|}, \quad (19)$$

$$\Phi_{12}(\phi) = \arg[U_1(\phi)U_2(\phi + \Delta_1)U_1^*(\phi + \Delta_2)U_2^*(\phi)]. \quad (20)$$

The overlap in M_{μ} is not a bare dot product of ED coefficient vectors, because the one-body Bloch basis itself changes with twist. For an electron configuration I , its coefficient overlap is weighted by

$$w_I = \prod_{p \in I} \langle u_p(\phi) | u_p(\phi + \Delta_{\mu}) \rangle. \quad (21)$$

For the nine-hole representation, the code uses the exactly

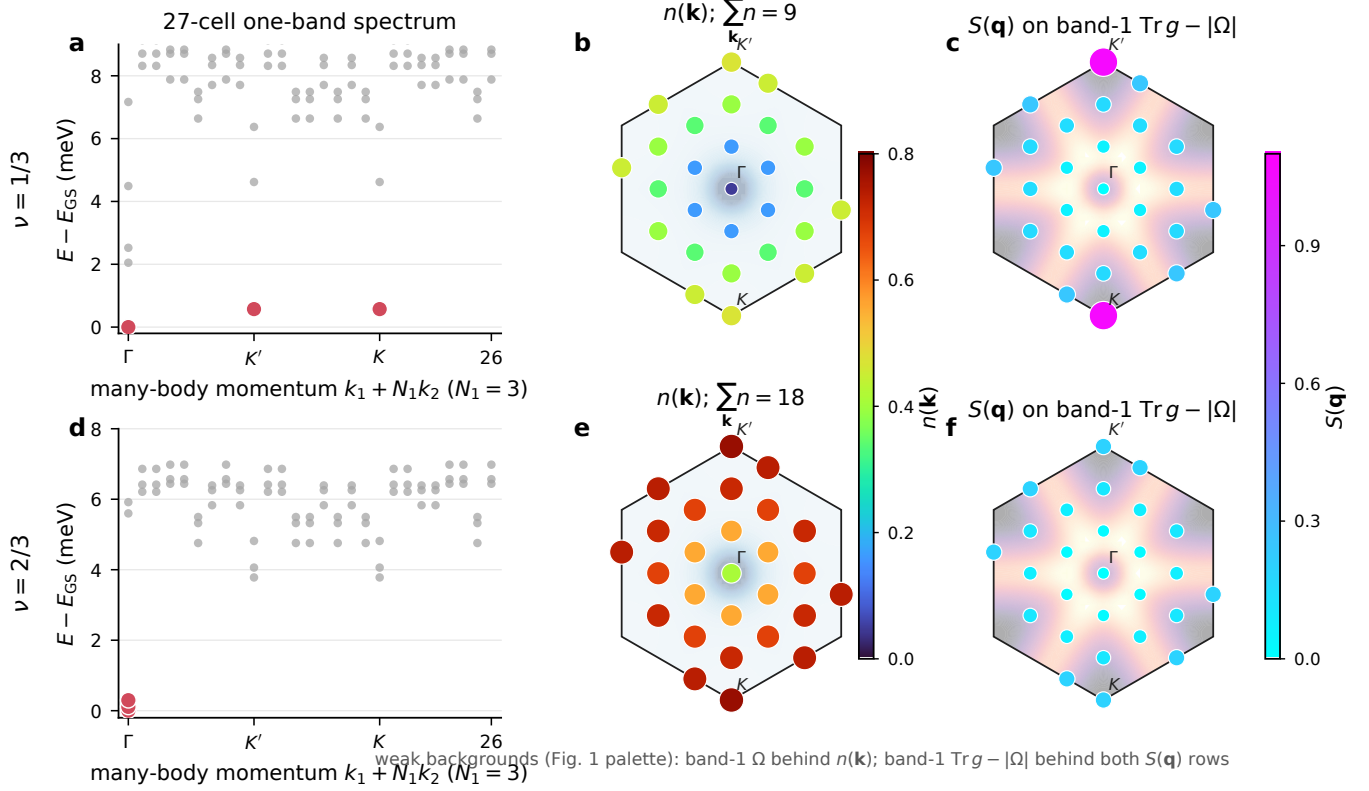


Figure 3: 27-site one-band reproduction of Supplementary Fig. 5. Top row: $\nu = 1/3$; bottom row: $\nu = 2/3$. (a,d) Momentum-resolved spectra, plotted as $E - E_{\text{GS}}$ in meV. Red markers denote the CDW or FCI three-state manifold. (b,e) Ground-manifold averaged $n(\mathbf{k})$ over weak, interpolated band-1 Berry curvature. (c,f) Projected $S(\mathbf{q})$ over the identical weak, interpolated band-1 $\text{Tr } g_1 - |\Omega_1|$ background. Bright marker colorbars encode the observables independently. The sharp zone-corner peaks at $\nu = 1/3$ and their absence at $\nu = 2/3$ distinguish the CDW and FCI responses.

equivalent electron complement,

$$w_H = \frac{\prod_{p=1}^{27} \langle u_p(\phi) | u_p(\phi + \Delta_\mu) \rangle}{\prod_{h \in H} \langle u_h(\phi) | u_h(\phi + \Delta_\mu) \rangle}. \quad (22)$$

The non-Abelian FHS curvature and discrete Chern number are

$$\begin{aligned} \mathcal{F}_{\phi_1 \phi_2} &= \frac{\Phi_{12}}{(\Delta\phi)^2}, \\ C_{\text{bundle}} &= \frac{1}{2\pi} \sum_{i,j} \Phi_{12}(\phi_{ij}), \\ \Delta\phi &= \frac{2\pi}{6}. \end{aligned} \quad (23)$$

We solve and cache the 7×7 grid endpoints as well as the 6×6 plaquettes. Links crossing a 2π boundary are consequently evaluated from endpoint Bloch overlaps, rather than by identifying raw ED eigenvectors. A separate analytic large-gauge sewing-matrix test is nevertheless still needed for a formal convergence claim. This single grid is a discrete diagnostic, not a twist-grid convergence study.

Figure 4 shows the two axial flux spectra and the curvature map. For $\nu = 1/3$, the bundle contains the lowest states from Γ, K, K' . Its Chern number is

$$C_{\text{bundle}}^{1/3} = 9.3 \times 10^{-7} \simeq 0, \quad (24)$$

and the smallest sampled separation between the three-state manifold and the nearest state in these three sectors is 1.476 meV. For $\nu = 2/3$, the three states all belong to Γ and

$$C_{\text{tracked}}^{2/3} = 1.99999975 \simeq 2. \quad (25)$$

If this tracked subspace is subsequently shown to be the globally isolated ground-state bundle, the conditional Hall response is $\sigma_{xy} = C_{\text{tracked}} e^2 / (3h) = 2e^2 / (3h)$. The minimum same- Γ separation is 5.144 meV, the minimum link singular value is 0.958 at $\nu = 2/3$, and the minimum tracked-sector overlap is 0.964 at $\nu = 1/3$. These checks show that the sampled links are nonsingular, but they do not prove isolation from the complete spectrum. Missing checks are the lowest relevant levels in all 27 translation sectors at every twist, the global $\min_\phi (E_4 - E_3)$, an explicit analytic 2π large-gauge sewing test, and $6 \times 6 \rightarrow 9 \times 9 \rightarrow 12 \times 12$ grid convergence.

7 ED benchmark protocol for QNS

The validated workflow is:

1. solve the plane-wave continuum Hamiltonian and verify shell convergence;

27-cell one-band ED under twisted boundaries ($\theta = 2.6^\circ$, $\varepsilon = 5$)

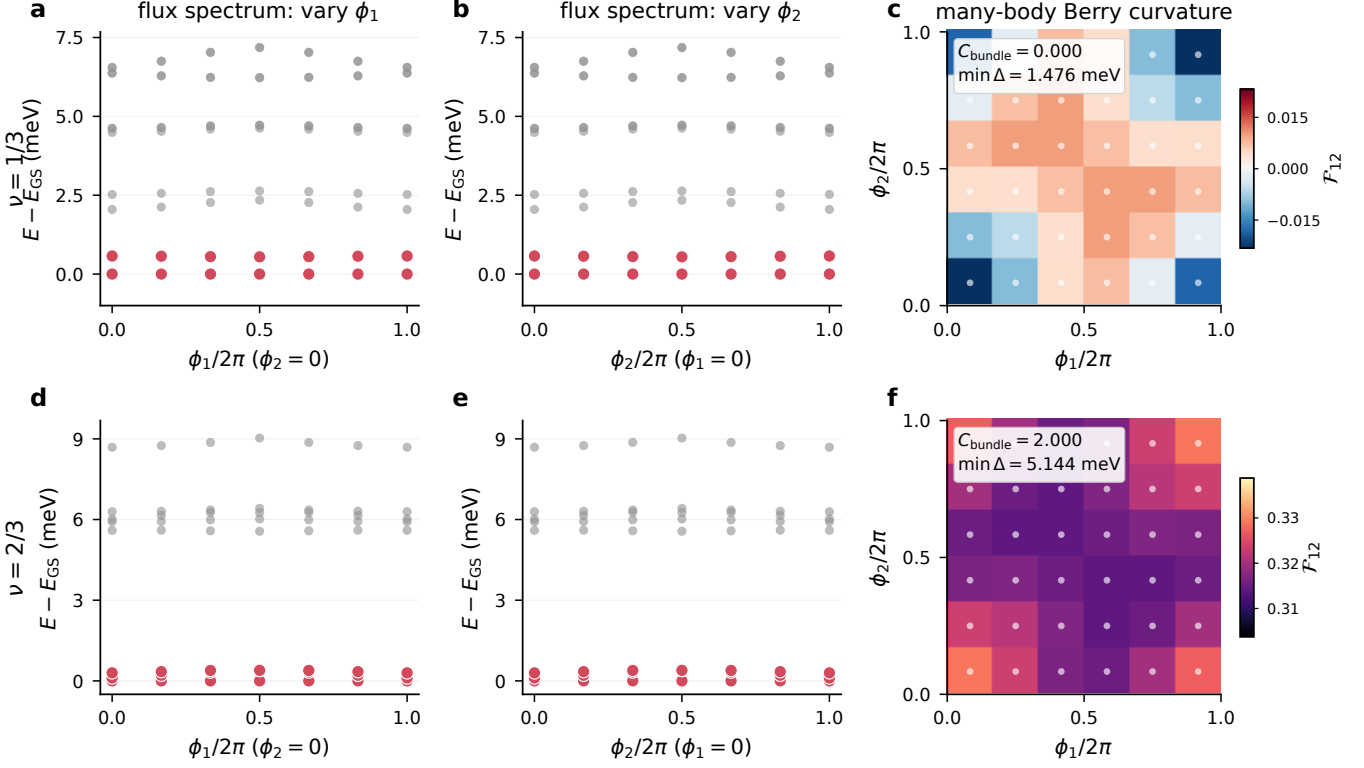


Figure 4: Twisted-boundary ED and many-body topology on the 27-cell one-band cluster. Top row: $\nu = 1/3$; bottom row: $\nu = 2/3$. (a,d) Spectral flow versus ϕ_1 at $\phi_2 = 0$. (b,e) Spectral flow versus ϕ_2 at $\phi_1 = 0$. At every twist, the lowest three energies are red scatter points. Gray points show five additional $\nu = 2/3$ levels and nine additional $\nu = 1/3$ levels sampled from the three CDW sectors. (c,f) Non-Abelian curvature $\mathcal{F}_{\phi_1\phi_2}$ on the original 6×6 primitive twist grid; white dots mark the FHS plaquette centers. The raw plaquette-flux sums give Chern number zero for the CDW and two for the tracked FCI manifold.

2. calculate band dispersion, Berry curvature and quantum metric on the same Bloch convention;
3. project the neutral-background Coulomb interaction with reciprocal-vector form factors;
4. fix charge- $U(1)$ and spin- $U(1)$, then block by translation; use point-group orbit equivalence at generic K , and split by C_3/C_6 character only when the rotation belongs to the little group of K ;
5. use sparse Hermitian Lanczos independently in each required sector;
6. compare energies, sector counting, normalized band weights, $n(\mathbf{k})$ and $S(\mathbf{q})$;
7. insert boundary flux, track the isolated ground manifold and integrate its non-Abelian Berry curvature;
8. only then benchmark the QNS energy, variance and state overlaps against ED.

Numerical regression checks support every stage. The general C_6 -supercell interaction agrees with the earlier rectangular 3×3 implementation to relative error 3.3×10^{-16} . Direct electron and particle-hole spectra agree

sector by sector to 3.4×10^{-13} meV. The finite plane-wave cutoff produces a maximum one-body C_6 mismatch of 8.0×10^{-6} meV, and the pre-symmetrization interaction Hermiticity error is 7.8×10^{-9} . The current complete 51-test regression suite passes; the topology outputs additionally record link conditioning, Lanczos residuals and interaction Hermiticity at every twist.

Part II

Neural many-body Bloch wavefunctions

8 Motivation and scope

The second calculation tests the neural many-body Bloch construction of Ref. [1] directly in the continuum model benchmarked in Part I. The purpose is not to fit coefficients in the one-band ED Hilbert space. Instead, a graph network dresses continuum Bloch orbitals in coordinate space and the resulting generalized Slater determinant is optimized against the complete interacting

continuum Hamiltonian. This can generate remote-band content even though the fixed one-body input is the mean-field first band.

This revision follows Appendix B, Eqs. (16)–(23), of Ref. [1]. In particular, the orbital transformation is the direct complex output

$$J_{\mathbf{k}i}(X) = \text{MLP}([\mathbf{k}, V_i^t]) \in \mathbb{C}, \quad Q_{\mathbf{k}i} = J_{\mathbf{k}i}\varphi_{\mathbf{k}i}. \quad (26)$$

There is no constant-one skip, residual parameterization, or factor 0.2; the implemented transformation is the direct $J_{\mathbf{k}i}$ above. The current source additionally implements three audit corrections: full- M uses an unbiased complex random initialization; the node aggregation is the literal $\sum_j m_{ij}$ in Eq. (19), including $j = i$; and the random auxiliary features h_i^0, h_{ij}^0 are explicitly retained in the trainable parameter tree. Earlier one-hot- M , no-self-message checkpoints are excluded from this Part. Figures 6, 7 and 8 and all result tables below use the four controlled 4,128-walker, 120-update baselines and the internal- and outer- C_3 extensions. C3-QNS was halted after 116 accepted updates when the persistent-chain energy had plateaued; each of the three outer sectors at both fillings completed 120 attempted updates. All quoted final energies come from independent random-start validation rather than the training chain.

9 Configuration, fixed Bloch basis, and conventions

A walker is a first-quantized configuration

$$X = \{(\mathbf{r}_i, l_i)\}_{i=1}^{N_e}, \quad \mathbf{r}_i \in [0, 1)^2, \quad l_i^{\text{code}} \in \{0, 1\}. \quad (27)$$

The fractional coordinate is converted to a Cartesian coordinate on the 3×3 torus. The layer input used by the network is

$$s_i = 2l_i^{\text{code}} - 1 \in \{-1, +1\}, \quad (28)$$

which is the $l_i = \pm 1$ convention of Ref. [1]. Particle labels are only row labels of the determinant and graph; no ordered occupation string is used.

The nine mesh momenta and plane-wave orbitals are

$$\mathbf{k}_{m_1 m_2} = \frac{m_1}{3}\mathbf{b}_1 + \frac{m_2}{3}\mathbf{b}_2, \quad m_1, m_2 = 0, 1, 2, \quad (29)$$

$$\phi_{\mathbf{k}}^0(\mathbf{r}, l) = \sum_{\mathbf{G}} u_{1\mathbf{k}\mathbf{G}}^l e^{i(\mathbf{k}+\mathbf{G})\cdot\mathbf{r}}. \quad (30)$$

The reciprocal set contains the same 61 $\mathbf{G}$ vectors per layer used for the single-particle bands in Fig. 1. The coefficients $u_{1\mathbf{k}\mathbf{G}}^l$ are obtained once by diagonalizing the continuum mean-field Hamiltonian. In JAX they are constants, and in PyTorch they are registered buffers. They are absent from the trainable parameter tree. Thus every run uses band 1 as its mean-field orbital source; neither ED amplitudes nor trainable plane-wave coefficients are used.

10 Appendix-B graph network

10.1 Initial physical and auxiliary features

The three reciprocal harmonics used at each vertex are $\mathbf{g}_a \in \{\mathbf{b}_1, \mathbf{b}_2, \mathbf{b}_1 + \mathbf{b}_2\}$. The immutable node feature is

$$v_i^0 = [\sin(\mathbf{g}_a \cdot \mathbf{r}_i), \cos(\mathbf{g}_a \cdot \mathbf{r}_i), s_i] \in \mathbb{R}^7. \quad (31)$$

For every ordered pair (i, j) , the immutable edge feature is

$$e_{ij}^0 = [\sin(\mathbf{k} \cdot (\mathbf{r}_i - \mathbf{r}_j)), \cos(\mathbf{k} \cdot (\mathbf{r}_i - \mathbf{r}_j)), s_i] \in \mathbb{R}^{19}, \quad (32)$$

where all nine mesh momenta are included. In particular, the edge layer channel is the source layer s_i , exactly as in Eq. (16) of Ref. [1]; no additional layer channels are appended.

The complete initial features are

$$V_i^0 = (v_i^0, h_i^0) \in \mathbb{R}^{32}, \quad E_{ij}^0 = (e_{ij}^0, h_{ij}^0) \in \mathbb{R}^{32}. \quad (33)$$

The code therefore uses 25 auxiliary node components and 13 auxiliary edge components. Appendix B explicitly calls h_i^0 and h_{ij}^0 “randomly initialized parameters.” They are therefore leaves of the trainable JAX pytree (and `nn.Parameter` objects in PyTorch), so natural-gradient updates act on them together with the other network parameters. The present code uses one random vector shared over particle labels and one shared over ordered pairs; sharing preserves permutation equivariance. Each component is initialized from $\mathcal{N}(0, 1/32)$. The paper does not specify this distribution or sharing rule, so both remain explicit local conventions. There is no learned input embedding before Eq. (36).

10.2 Two message-passing iterations

For iteration $a = 1, 2$,

$$Q_{ij}^a = W_Q^a E_{ij}^{a-1}, \quad (34)$$

$$K_{ij}^a = W_K^a E_{ij}^{a-1}, \quad (35)$$

$$m_{ij}^a = F^a \left(\sum_n Q_{in}^a \otimes K_{nj}^a \right) \odot G^a(E_{ij}^{a-1}), \quad (36)$$

$$h_i^a = h_i^{a-1} + U^a \left(\left[\sum_j m_{ij}^a, V_i^{a-1} \right] \right), \quad (37)$$

$$h_{ij}^a = h_{ij}^{a-1} + H^a([m_{ij}^a, E_{ij}^{a-1}]), \quad (38)$$

$$V_i^a = (v_i^0, h_i^a), \quad (39)$$

$$E_{ij}^a = (e_{ij}^0, h_{ij}^a). \quad (40)$$

The outer product in Eq. (36) is flattened before F^a . F^a, G^a, H^a, U^a are one-layer real affine maps followed by exact GELU. The two iterations have independent parameters. No division by N_e , no neighbor-count normalization, and no LayerNorm is applied. Self edges remain available to both the edge update and node aggregation.

This is the literal $\sum_j$ printed in Luo–Fu Eq. (19), rather than the “all other particles” neighborhood convention of Ref. [3]. The literal physical features v_i^0, e_{ij}^0 are reinserted after every iteration.

11 Backflow, orbital transformation, and determinant

The complex backflow matrix is implemented as a real, bias-free $32 \rightarrow 4$ linear map. Its first and last two outputs are the real and imaginary parts of a two-component displacement,

$$\delta \mathbf{r}_i(X) = WV_i^2, \quad \tilde{\mathbf{r}}_i = \mathbf{r}_i + \delta \mathbf{r}_i. \quad (41)$$

There is no tanh, clipping, or external backflow scale. The fixed mean-field coefficients are evaluated only at this transformed coordinate,

$$\varphi_{\mathbf{k}i}(X) = \sum_{\mathbf{G}} u_{1\mathbf{k}\mathbf{G}}^{l_i} \exp[i(\mathbf{k} + \mathbf{G}) \cdot (\mathbf{r}_i + \delta \mathbf{r}_i(X))]. \quad (42)$$

Consequently, backflow does not update $u_{1\mathbf{k}\mathbf{G}}^{l_i}$. The factors $\exp[i(\mathbf{k} + \mathbf{G}) \cdot \delta \mathbf{r}_i]$ are generated only by changing the coordinate argument.

The neural orbital transformation uses the Cartesian two-component mesh momentum and the final node feature,

$$[V_i^2, k_x, k_y] \in \mathbb{R}^{34} \xrightarrow{34 \rightarrow 32} \text{GELU} \xrightarrow{32 \rightarrow 2} (J_{\mathbf{k}i}^{\text{R}}, J_{\mathbf{k}i}^{\text{I}}), \quad (43)$$

$$J_{\mathbf{k}i} = J_{\mathbf{k}i}^{\text{R}} + iJ_{\mathbf{k}i}^{\text{I}}, \quad Q_{\mathbf{k}i} = J_{\mathbf{k}i} \varphi_{\mathbf{k}i}. \quad (44)$$

The direct complex output is not converted to $1 + \alpha \Delta J$. Equation (44) is implemented once: J is not included inside φ and then multiplied again. A unit test that sets $J_{\mathbf{k}i} = 1/2$ verifies $\det Q = 2^{-N_e} \det \varphi$ to numerical precision.

For the Luo–Fu full- M state,

$$\Phi_{\gamma i}(X) = \sum_{\mathbf{k}} M_{\gamma \mathbf{k}} Q_{\mathbf{k}i}(X), \quad (45)$$

$$\Psi_M(X) = \det_{\gamma i} \Phi_{\gamma i}(X), \quad M \in \mathbb{C}^{N_e \times 9}. \quad (46)$$

All graph, W , J , and complex M parameters are trainable. Only one final generalized Slater determinant is used. Because M is trainable and dense, the Cauchy–Binet expansion of $\det(MQ)$ generally contains several total-momentum sectors.

The no- M ablation removes the matrix rather than replacing it by another trainable map,

$$\Psi_{\Gamma}(X) = \det_{\alpha i} Q_{\mathbf{k}\alpha i}(X), \quad \sum_{\alpha=1}^{N_e} \mathbf{k}_{\alpha} = \Gamma \pmod{\mathbf{G}}. \quad (47)$$

For $N_e = 3$, the selected fractional momenta are $\{(2,0), (2,1), (2,2)\}/3$; for $N_e = 6$ they are $\{(1,0), (1,1), (1,2), (2,0), (2,1), (2,2)\}/3$. They minimize

the sum of bare band-1 energies subject to total momentum Γ . Each generalized orbital retains its own single-particle momentum; only the determinant has total momentum zero. No C_3 label or ED rotation eigenvalue is supplied.

12 Internally equivariant C3-QNS extension

The periodic Fourier encoding in Eqs. (31) and (32) guarantees torus periodicity and primitive-translation symmetry, as intended in Ref. [1]. It does *not* by itself impose C_3 equivariance: the three node harmonics close under rotation, but a generic learned linear vector head need not transform as a vector, and the nine continuous edge harmonics are not a closed regular representation. The present C3-QNS therefore keeps the Luo–Fu message-passing network but constrains its use by a Reynolds average.

Let $g = C_{3,A}$ be the $-2\pi/3$ active rotation about the reference stacking center A , with Cartesian matrix R . For a raw network output,

$$\delta \mathbf{r}_i^{C_3}(X) = \frac{1}{3} \sum_{p=0}^2 R^{-p} \delta \mathbf{r}_i^{\text{raw}}(g^p X), \quad (48)$$

$$J_{\mathbf{k}i}^{C_3}(X) = \frac{1}{3} \sum_{p=0}^2 J_{m^p \mathbf{k}, i}^{\text{raw}}(g^p X), \quad (49)$$

where m is the discrete one-band momentum permutation induced by g . The same network parameters are used in all three terms. These definitions give $\delta \mathbf{r}_i^{C_3}(gX) = R \delta \mathbf{r}_i^{C_3}(X)$ and the corresponding covariance of J .

A gauge correction is required after backflow. In the plane-wave convention of Part I, the bare first-band spinor obeys

$$\phi_{m\mathbf{k}}^0(g\tilde{\mathbf{r}}, l) = z_{\mathbf{k}} e^{-i\mathbf{a}_l \cdot \tilde{\mathbf{r}}} \phi_{\mathbf{k}}^0(\tilde{\mathbf{r}}, l), \quad \tilde{\mathbf{r}} = \mathbf{r} + \delta \mathbf{r}, \quad (50)$$

where $z_{\mathbf{k}} \in U(1)$ is extracted from the continuum eigenvectors and $\mathbf{a}_l$ is the reciprocal-momentum shift required to return the rotated plane-wave expansion to the code’s layer-dependent Bloch gauge. In the present convention,

$$\mathbf{a}_0 = \mathbf{b}_1, \quad \mathbf{a}_1 = -\mathbf{b}_2. \quad (51)$$

This is the sense in which a *momentum compensation* is required. It does not alter the mesh momentum $\mathbf{k}$, mix different $\mathbf{k}$ values, or update the fixed coefficients $u_{1\mathbf{k}\mathbf{G}}^{l_i}$.

To see the mismatch explicitly, backflow equivariance gives

$$\tilde{\mathbf{r}}_i(gX) = R\mathbf{r}_i + R\delta \mathbf{r}_i(X) = R\tilde{\mathbf{r}}_i(X). \quad (52)$$

Substitution into Eq. (50) then gives

$$\begin{aligned} \phi_{m\mathbf{k}}^0(\tilde{\mathbf{r}}_i(gX), l_i) &= z_{\mathbf{k}} e^{-i\mathbf{a}_{l_i} \cdot [\mathbf{r}_i + \delta \mathbf{r}_i(X)]} \phi_{\mathbf{k}}^0(\tilde{\mathbf{r}}_i(X), l_i) \\ &= z_{\mathbf{k}} e^{-i\mathbf{a}_{l_i} \cdot \mathbf{r}_i} \underbrace{e^{-i\mathbf{a}_{l_i} \cdot \delta \mathbf{r}_i(X)}}_{\eta_i(X)} \phi_{\mathbf{k}}^0(\tilde{\mathbf{r}}_i(X), l_i). \end{aligned} \quad (53)$$

The physical continuum C_3 action contains the first phase, evaluated at the actual electron coordinate $\mathbf{r}_i$. The second factor appears only because the fixed Bloch orbital was evaluated at the neural coordinate.

Define $h_l(\delta\mathbf{r}) = \exp(i\mathbf{b}_l \cdot \delta\mathbf{r})$. Cancellation of the second factor requires

$$\frac{h_l(R\delta\mathbf{r})}{h_l(\delta\mathbf{r})} = e^{i\mathbf{a}_l \cdot \delta\mathbf{r}}. \quad (54)$$

Since $h_l(R\delta\mathbf{r})/h_l(\delta\mathbf{r}) = \exp\{i[(R^\top - I)\mathbf{b}_l] \cdot \delta\mathbf{r}\}$, we solve the fixed two-dimensional linear system

$$(R^\top - I)\mathbf{b}_l = \mathbf{a}_l \quad (55)$$

once and form

$$\bar{Q}_{\mathbf{k}i}(X) = e^{i\mathbf{b}_{l_i} \cdot \delta\mathbf{r}_i^{C_3}(X)} J_{\mathbf{k}i}^{C_3}(X) \phi_{\mathbf{k}}^0(\mathbf{r}_i + \delta\mathbf{r}_i^{C_3}(X), l_i). \quad (56)$$

Equations (53)–(56) imply

$$\bar{Q}_{m\mathbf{k},i}(gX) = z_{\mathbf{k}} e^{-i\mathbf{a}_{l_i} \cdot \mathbf{r}_i} \bar{Q}_{\mathbf{k}i}(X). \quad (57)$$

For a many-body minor, the product of the physical phases is

$$G_A(X)^{-1} = \exp\left[-i \sum_i \mathbf{a}_{l_i} \cdot \mathbf{r}_i\right], \quad (58)$$

while the product of $z_{\mathbf{k}}$ and row-sorting parity gives the configuration-independent sewing number ζ_p used below.

The dot products are bilinear, with no complex conjugation. Because Luo–Fu backflow is complex, the compensation can change both phase and amplitude; it is fixed analytically by covariance and adds no trainable parameter. In code, $\mathbf{b}_l$ is constructed once during initialization and the column factor is applied at `src/jax_neural_bloch.py:173–177`; the independent PyTorch path is `src/neural_bloch.py:494–503`. Omitting it leaves an order-one, configuration-dependent character error despite an equivariant displacement.

Let $S_0 = \{\mathbf{k}_\alpha\}$ be the same lowest-energy total- Γ subset as the no- M baseline and $S_p = m^p S_0$. Sorting each minor gives

$$\zeta_p = \text{sgn}(m : S_p \rightarrow S_{p+1}) \prod_{\mathbf{k} \in S_p} z_{\mathbf{k}}. \quad (59)$$

For requested character $\chi_m = \exp(2\pi i m/3)$,

$$c_0 = 1, \quad c_{p+1} = \frac{\chi_m}{\zeta_p} c_p, \quad c_3 = c_0, \quad (60)$$

and the final ansatz is

$$\Psi_{C_3,m}(X) = \frac{1}{3} \sum_{p=0}^2 c_p \det[\bar{Q}_{\mathbf{k}i}(X)]_{\mathbf{k} \in S_p}. \quad (61)$$

This is a constrained realization of the Luo–Fu determinant index w : the three minors are one symmetry orbit with shared neural parameters and fixed sewing weights, rather than three independently optimized determinants.

The full- M matrix is absent, so total momentum remains Γ .

Only primitive translations and $C_{3,A}$ are independent generators. The other centers satisfy $C_{3,B} = T_{\mathbf{t}_{BA}} C_{3,A}$ and $C_{3,C} = T_{\mathbf{t}_{CA}} C_{3,A}$. Translation symmetry plus Eq. (48) therefore implies the B/C relations; all three centers remain useful regression tests. On random configurations the JAX/PyTorch log-wavefunction mismatch is 1.7×10^{-14} . For the frozen $\nu = 1/3$, $m = 0$ checkpoint, a 512-configuration independent test gives a maximum complete-state character residual 7.12×10^{-5} (95th percentile 1.77×10^{-5}), consistent with the finite 61-plane-wave truncation. Thus the constraint is numerically satisfied; its energetic value remains a separate question.

13 Complete-wavefunction outer C_3 projection

The second symmetry construction deliberately removes the internal Reynolds averages in Eqs. (48)–(49). Its raw state is exactly the periodic no- M , total- Γ generalized determinant

$$D_\theta(X) = \det[Q_{\mathbf{k}_\alpha i}(X)]_{\mathbf{k}_\alpha \in S_0}, \quad (62)$$

with an unrestricted Luo–Fu message-passing backflow and J network. Thus neither an individual displacement nor an individual generalized orbital is forced to be a C_3 eigenobject.

The physical continuum action includes the layer-dependent Bloch-gauge sewing. With the code’s active-rotation convention, define

$$(C^a D_\theta)(X) = G_a(X) D_\theta(C_3^a X), \quad (63)$$

where $G_0 = 1$ and G_a is the accumulated product of the physical layer-gauge factor at each preceding point on the configuration orbit. For $\omega = \exp(2\pi i/3)$, the complete-state projector is

$$\boxed{\begin{aligned} \Psi_{m,\theta}(X) &= P_m D_\theta(X) \\ &= \frac{1}{3} \sum_{a=0}^2 \omega^{-ma} G_a(X) \\ &\quad \times D_\theta(C_3^a X), \quad m = 0, 1, 2. \end{aligned}} \quad (64)$$

The three independently optimized sectors have the same 78,100 real parameters as no M ; projection adds no trainable parameter. Every raw branch is a complete graph/determinant evaluation with shared parameters.

Crucially, all stochastic quantities use Eq. (64), not one raw branch. Metropolis ratios are $|\Psi_m(X')/\Psi_m(X)|^2$, every kinetic or tunnelling local-energy term contains the complete ratio $\Psi_m(X')/\Psi_m(X)$, and SR uses $\mathcal{O}_\mu(X) = \partial_{\theta_\mu} \log \Psi_m(X)$. Setting `outer_c3_projector=False` recovers the no- M wavefunction exactly.

Before production, 64 random configurations gave relative projection norms 0.5411, 0.5929 and 0.6410 for

$m = 0, 1, 2$, so no sector was annihilated. The maximum character errors in $\Psi_m(C_3X)/\Psi_m(X) = \omega^m$ were 6.13×10^{-14} , 6.39×10^{-14} and 5.02×10^{-14} . All idempotency/orthogonality residuals for $P_m P_n = \delta_{mn} P_m$ were below 7.55×10^{-15} , and disabling the projector changed the no- M output by exactly zero. These tests isolate the new outer-projection logic from the

14 Initialization and trainable parameter count

Every affine neural-network kernel uses JAX LeCunNormal, $\text{std} = 1/\sqrt{\text{fan in}}$, with zero biases. The auxiliary h^0 convention was specified above. For full- M , the real and imaginary components are independently initialized with half-scale LeCunNormal,

$$\mathbb{E}[(\Re M_{\gamma k})^2] = \mathbb{E}[(\Im M_{\gamma k})^2] = \frac{1}{2N_k}, \quad \mathbb{E}|M_{\gamma k}|^2 = \frac{1}{N_k}. \quad (65)$$

Thus every momentum minor is accessible at step zero and there is no preferred Γ subset. Ref. [1] states global LeCunNormal network initialization but does not separately specify M ; the complex extension and variance split above are therefore the present user-requested convention, not an attributed statement from the paper.

The mean-field orbital source is always band 1. However, because W and J themselves are LeCunNormal rather than forced to $W = 0, J = 1$, the step-zero many-body state is not claimed to equal the undressed bare Slater determinant pointwise. This distinction is recorded to make the initialization auditable.

Table 3: Trainable real parameter counts of the strict one-determinant implementation.

filling	N_e	full M	no M, Γ	internal C3	outer P_m
1/3	3	78,154	78,100	78,100	78,100
2/3	6	78,208	78,100	78,100	78,100

The difference is exactly $2N_e N_k$ real parameters for the complex $N_e \times 9$ matrix M . The internal Reynolds average and the external P_m projector share the raw no- M parameters among all three rotated branches, so neither symmetry construction adds a trainable parameter.

15 Hamiltonian and local-energy estimator

The VMC objective uses the complete continuum Hamiltonian at $\theta = 3^\circ$, screening parameter $\epsilon = 5$, and a 3×3 torus. It is not the one-band or five-band projected ED Hamiltonian. The local energy contains the kinetic term, layer-dependent moiré potential, spatially varying inter-

layer tunnelling, and periodic Coulomb interaction,

$$E_{\text{loc}}(X) = \frac{(H\Psi)(X)}{\Psi(X)} = E_{\text{kin}} + E_{\text{moiré}} + E_{\text{tun}} + E_C. \quad (66)$$

The Coulomb sum uses reciprocal-shell cutoff 8, dielectric $\epsilon = 5$, and removes the $\mathbf{q} = 0$ neutral-background component. Tunnelling is evaluated by flipping one layer index and multiplying the exact wavefunction ratio.

Cartesian derivatives of the complete complex state use the three-point central stencil

$$\partial_a \Psi / \Psi \simeq \frac{\Psi(X + h\hat{a}) - \Psi(X - h\hat{a})}{2h\Psi(X)}, \quad (67)$$

$$\partial_a^2 \Psi / \Psi \simeq \frac{\Psi(X + h\hat{a}) + \Psi(X - h\hat{a}) - 2\Psi(X)}{h^2\Psi(X)}. \quad (68)$$

Here $h = 1.5 \times 10^{-3} a_M$. Periodic wrapping is applied after every coordinate displacement. A previous $h, h/2, h/4$ audit is not reused as a validation of the new checkpoints; it must be repeated after convergence.

15.1 Raw-coordinate and full-wavefunction Hamiltonian audit

The production JAX path is `src/jax_neural_block.py:184-248`; the independent PyTorch reference is `src/continuum_vmc.py:96-206`. Both implementations first form $\mathbf{r}_i$ from the input *raw positions*. Consequently,

$$V_M(X) = -2V \sum_{i,a} \cos(\mathbf{g}_a \cdot \mathbf{r}_i + \phi_{l_i}), \quad (69)$$

$$V_C(X) = \sum_{i < j} \sum_{\mathbf{q} \neq 0} v_{\mathbf{q}} \cos[\mathbf{q} \cdot (\mathbf{r}_i - \mathbf{r}_j)] \quad (70)$$

never see $\delta \mathbf{r}_i$. For tunnelling, $t(\mathbf{r}_i)$ is evaluated at that same raw coordinate, while its numerator is the complete $\Psi(X_{l_i \rightarrow 1-l_i})$ obtained by a new `logpsi` call. The kinetic ratios similarly use new complete calls,

$$R_{i\mu}^\pm = \frac{\Psi(X_{\mathbf{r}_i \rightarrow \mathbf{r}_i \pm h\hat{\mu}})}{\Psi(X)}, \quad (71)$$

so each displaced call rebuilds the graph and recomputes every particle's $\delta \mathbf{r}_j$, $J_{\mathbf{k}j}$, the MQ contraction, and the determinant. No backflow displacement is cached by the Hamiltonian.

As a final energy-zero and convention check, a fixed-state regression sets $\delta \mathbf{r} = 0$, $J = 1$, and a fixed M selecting a known two-orbital bare Bloch Slater determinant. The four local-energy components are then compared with an independent finite-basis evaluation. This test uses 4,096 walkers, 120 burn sweeps and 10 measurements and is never used for an optimization update.

Table 4: Fixed-state Hamiltonian regression in meV for the complete many-body state. Parentheses are Monte Carlo standard errors.

term	finite-basis	Monte Carlo	difference
kinetic	39.4207	39.3467(3343)	-0.0740
moiré	-38.5168	-38.7739(5347)	-0.2571
tunnelling	-29.9435	-29.6118(2984)	+0.3317
Coulomb	-9.6769	-10.1671(3354)	-0.4902
total	-38.7164	-39.2060(3354)	-0.4896

Every component and the total agree within 1.5 quoted standard errors. The regression therefore finds no common Hamiltonian normalization, Coulomb background, tunnelling, or energy-zero mismatch at its present precision.

A targeted regression test in `tests/test_jax_neural_bloch.py` instruments the real `_encoded_graph` function. One local-energy call produces four complete graph evaluations: base X with shape $(B, N_e, 2)$, all plus and all minus configurations with shape $(2N_e B, N_e, 2)$ each, and all one-particle layer flips with shape $(N_e B, N_e, 2)$. It checks every displaced coordinate and every flipped layer against an independently constructed array. The test passes, as does the JAX-PyTorch full-local-energy parity test to 2×10^{-7} meV absolute tolerance. No Hamiltonian input-routing error was found in this audit.

16 MCMC and natural-gradient optimization

There are 4,128 simultaneous walkers. Initial fractional coordinates are uniform on the torus and layer indices are uniform in $\{0, 1\}$. Each sweep proposes, for every walker, one random-particle Gaussian displacement with standard deviation 0.06 in fractional supercell coordinates, followed by one random-particle layer flip. Both use the Metropolis ratio $\min[1, |\Psi(X')/\Psi(X)|^2]$. The present jobs use 300 burn-in sweeps and two sweeps per update. Acceptance for coordinate and layer moves is recorded separately.

For real parameters θ_a ,

$$O_a(X) = \partial_{\theta_a} \log \Psi_{\theta}(X), \quad (72)$$

$$S_{ab} = \text{Re}[\langle O_a^* O_b \rangle - \langle O_a^* \rangle \langle O_b \rangle], \quad (73)$$

$$F_a = 2 \text{Re} \langle (E_{\text{loc}} - \langle E_{\text{loc}} \rangle) O_a^* \rangle. \quad (74)$$

The update is

$$(S + \lambda I) \Delta \theta = F, \quad \theta \leftarrow \theta - \eta \Delta \theta, \quad (75)$$

with ordinary, unpreconditioned CG. JAX JVP/VJP operations apply the metric matrix-free. Starting at iteration 20 and after every subsequent Krylov step, the code explicitly recomputes the authoritative residual

$$r_{\text{rel}}^{\text{true}} = \frac{\|F - (S + \lambda I) \Delta \theta\|_2}{\|F\|_2} \quad (76)$$

with an additional Fisher-vector product. The solve stops immediately when $r_{\text{rel}}^{\text{true}} < 0.02$ or at 60 CG iterations. At the cap, an update is accepted only if $r_{\text{rel}}^{\text{true}} < 0.05$ and $F^T \Delta \theta > 0$. Otherwise the parameter update is rejected and the following step uses the recovery values $\lambda = 0.03$ and $\eta = 10^{-3}$. After two consecutive stable recovery solves, the protocol returns to $\lambda = 0.01$ and $\eta = 2 \times 10^{-3}$. Every trace stores the true residual, $F^T \Delta \theta$, force and direction norms, acceptances, and whether the update was applied. This is an adaptive truncated natural gradient, not a claim that the 10^{-3} linear-system tolerance has been reached.

Table 5: Optimization and numerical settings. Quantities not specified by Ref. [1] are explicitly marked as local choices.

quantity	current value	provenance
width / message iterations	32 / 2, unshared	Ref. [1], Appendix B
orbital MLP hidden width	32, GELU	Ref. [1], Appendix B
final determinants	1	local controlled choice
optimizer / implementation	natural gradient / JAX 0.11.1	Ref. [1]
learning rate	2×10^{-3}	Ref. [1]
walkers per update	4,128	Ref. [1]
affine-kernel initialization	LeCunNormal	Ref. [1]
h^0 initialization	random, trainable	random: Ref. [1]; law/sharing: local
full- M initialization	complex LeCunNormal	user choice; paper unspecified
node aggregation	$\sum_j m_{ij}$, including self	Ref. [1], Eq. (19)
updates in present experiment	120	user-specified finite run
paper long-run statement	1,000 for clusters $> 3 \times 3$	Ref. [1]
burn / sweeps per update	300 / 2	local choice
proposal scale	0.06 fractional	local choice
SR damping	10^{-2}	local choice
CG range / true-residual rules	20-60 / 0.02 stop, 0.05 accept	user-specified
wavefunction / SR batch	516 / 516	local memory choice
local-energy batch	32	local memory choice
finite-difference step	$1.5 \times 10^{-3} a_M$	local choice
random seed	83	matched local choice
precision	JAX 64 bit	local numerical choice

All 4,128 walkers are used in the energy, force, and metric estimate at each update. The printed within-step SEM treats walker values as an instantaneous ensemble diagnostic; it is not an independent validation error. Checkpoints at initialization, every 10 updates, the latest completed update, and update 120 store both flattened parameters, optimizer recovery state, sampler RNG, and walker coordinates/layers. Interrupted jobs resume this complete state; no validation configuration ever enters an update.

17 Implementation checks completed before production

The current complete 51-test suite passes; selected architecture, parity, observable, and local-energy checks verify:

1. particle exchange changes the determinant sign and no other factor;
2. $J_{\mathbf{k}i}$ is applied exactly once;
3. fixed Bloch coefficients are absent from trainable parameters;
4. shifted-coordinate Bloch evaluation agrees with an explicit plane-wave calculation;
5. PyTorch and JAX $\log \Psi$ agree at identical parameters;

6. complete JAX and PyTorch local energies agree;
7. no- M is invariant under both primitive translations and contains no trainable M array;
8. the complete-state C_3 projector satisfies character, idempotency, orthogonality, nonzero-sector-norm and projector-disabled no- M regression tests before its three-sector production run;
9. matrix-free force and Fisher-vector products agree with an explicit score-Jacobian calculation;
10. checkpoint export between JAX and PyTorch is lossless;
11. full- M is reproducibly random, complex, and not one-hot;
12. h_i^0, h_{ij}^0 are nonzero trainable parameters; and
13. a one-particle graph receives its nonzero $j = i$ message in Eq. (19);
14. the 1-RDM uses full replaced-configuration ratios and a fixed bare projector, and its bare-Slater full-basis trace is N_e ; and
15. kinetic displacements and tunnelling flips rebuild the complete JAX graph, while diagonal Hamiltonian factors use raw coordinates.

A one-update end-to-end smoke test also passed MCMC, local-energy evaluation, truncated SR, residual recording, parameter update, and checkpoint writing. These checks validate implementation consistency, not physical convergence.

18 Controlled baseline and C3-QNS experiments

Four baseline jobs were trained from independent random walker ensembles with the same protocol. Full M has a dense trainable complex momentum matrix and therefore mixes many-body momentum sectors; no M removes this matrix and uses a fixed orbital set with total momentum Γ . A fifth job adds internally equivariant C3-QNS at $\nu = 1/3$. It was stopped at accepted update 116 after the requested plateau check. Table 6 lists the result directories; the frozen C3 checkpoint is `jax_neural_bloch_step_0116.npz`. Three additional $\nu = 1/3$ jobs use the unrestricted no- M raw network with the complete-wavefunction projector P_m , $m = 0, 1, 2$. Each completed 120 updates with 4,128 persistent walkers; all three frozen outer-projected The identical three-sector protocol was subsequently run at $\nu = 2/3$. Sectors are optimized independently after the common initialization and are selected only with fresh, re-equilibrated validation chains. checkpoints are `jax_neural_bloch_step_0120.npz`.

Table 6: Authoritative calculations used in this report. C3-QNS was stopped at update 116; the four baselines and six outer sectors completed 120 updates.

ν	ansatz	K_{tot}	result-directory suffix
1/3	full M	mixed	<code>adaptive_cg120_20260821_full_m_nu1of3</code>
1/3	no M	Γ	<code>adaptive_cg120_20260821_no_m_gamma_nu1of3</code>
1/3	C3-QNS	$\Gamma, m = 0$	<code>c3_qns_adaptive120_nu1of3</code>
1/3	outer P_0	$\Gamma, m = 0$	<code>outer_c3_projector_nu1of3_120/m0</code>
1/3	outer P_1	$\Gamma, m = 1$	<code>outer_c3_projector_nu1of3_120/m1</code>
1/3	outer P_2	$\Gamma, m = 2$	<code>outer_c3_projector_nu1of3_120/m2</code>
2/3	outer P_0	$\Gamma, m = 0$	<code>outer_c3_projector_nu2of3_120/p0</code>
2/3	outer P_1	$\Gamma, m = 1$	<code>outer_c3_projector_nu2of3_120/p1</code>
2/3	outer P_2	$\Gamma, m = 2$	<code>outer_c3_projector_nu2of3_120/p2</code>
2/3	full M	mixed	<code>adaptive_cg120_20260821_full_m_nu2of3</code>
2/3	no M	Γ	<code>adaptive_cg120_20260821_no_m_gamma_nu2of3</code>

18.1 Training-chain diagnostics

Table 7 summarizes the final ten attempted updates for each frozen trace (steps 107–116 for C3-QNS). The mean and RMS describe the persistent training chain and are not independent energy estimates. A rejected update contributes its sampled local energy to the diagnostic trace but leaves the parameters unchanged. The two $\nu = 1/3$ jobs never activated the rejection rule. At $\nu = 2/3$, both jobs rejected 19 of 120 attempted updates; the final full- M attempt was rejected because $r_{\text{rel}}^{\text{true}} = 0.0512$ exceeded the 0.05 cap. The saved step-120 checkpoint therefore contains the parameters after the last accepted update, not the rejected direction.

Table 7: Final-ten training diagnostics. Energies are meV per particle; RMS is training-chain scatter.

ν	ansatz	$\langle E/N_e \rangle$	min. E/N_e	RMS	$\langle r_{\text{rel}} \rangle$	$\langle N_{\text{CG}} \rangle$	reject ₁₂₀
1/3	full M	-35.9952	-36.0866	0.0434	0.0183	48.0	0
1/3	no M, Γ	-36.7708	-36.9023	0.0592	0.0201	56.7	0
1/3	C3-QNS, Γ	-37.1749	-37.2188	0.0287	0.0163	37.7	0
1/3	outer P_0	-37.1660	-37.2401	0.0532	0.0187	34.1	0
1/3	outer P_1	-32.6293	-32.6686	0.0319	0.0181	50.5	0
1/3	outer P_2	-35.9573	-36.0274	0.0371	0.0179	48.4	0
2/3	outer P_0	-52.4662	-52.5856	0.0484	0.0745	50.8	17
2/3	outer P_1	-51.5899	-51.7154	0.0805	0.0385	55.6	23
2/3	outer P_2	-52.7891	-52.8479	0.0581	0.0202	53.4	8
2/3	full M	-52.0575	-52.1563	0.0817	0.0371	54.5	19
2/3	no M, Γ	-52.4147	-52.4842	0.0578	0.0397	54.9	19

The final-ten mean position/layer acceptances are 0.681/0.592, 0.669/0.618 and 0.672/0.643 for the $\nu = 1/3$ full- M , no- M and C3-QNS states, and 0.651/0.604 and 0.660/0.632 at $\nu = 2/3$. The corresponding At $\nu = 2/3$, outer P_0, P_1, P_2 have final-ten position/layer acceptances 0.651/0.622, 0.663/0.641 and 0.654/0.624, with mean variances 681.3, 703.9 and 684.8 meV². Their cumulative rejected-update counts are 17, 23 and 8, respectively. final-ten local-energy variances are 118.0, 104.5, 90.2, 701.0 and 718.3 meV². Thus a smooth energy trace does not establish a zero-variance eigenstate. For outer P_0, P_1, P_2 , the corresponding final-ten position/layer acceptances are 0.671/0.633, 0.695/0.584 and 0.671/0.526; the mean variances are 77.7, 120.2 and 99.2 meV². All three outer runs accepted all 120 updates.

18.2 Independent validation and checkpoint selection

Training walkers are never used to select the reported final energy. After the step-120 checkpoints were frozen, each candidate set was tested with 1,024 new random walkers, 200 burn-in sweeps and ten measurement sweeps. A local energy was evaluated after every measurement sweep, giving 10,240 measurements per state. Within each comparison set, the lower re-equilibrated low-cost energy selects the one state that receives a second validation with 4,128 new walkers, 300 burn-in sweeps and 20 measured sweeps (82,560 local energies). The three outer sectors are selected separately at each filling. The selection is therefore independent of the minimum training-chain point; in particular, the $\nu = 2/3$ training-chain ordering is explicitly checked rather than trusted. The stopped C3-QNS checkpoint was separately tested with the same low-cost and precise protocols, so its comparison to no M uses equal-size, independently re-equilibrated precise ensembles.

Table 8: Independent validation energies in meV per particle. Parentheses are eight-block, sweep-major standard errors. ΔE_{5b} uses the matching five-band ED energy.

ν	ansatz	low-cost E/N_e	precise E/N_e	variance	ΔE_{5b}
1/3	full M	-36.14253(4822)	—	113.90	+1.25071
1/3	no M , Γ	-36.79290(2589)	-36.78317(1019)	120.65	+0.61006
1/3	C3-QNS, Γ	-36.28392(10963)	-36.71907(1850)	1721.85	+0.67416
1/3	outer P_0	-37.14657(3297)	-37.19495(1053)	81.90	+0.19828
1/3	outer P_1	-32.60137(3134)	—	131.50	+4.79186
1/3	outer P_2	-35.91579(3900)	—	103.01	+1.47744
2/3	full M	-52.01665(4485)	—	730.94	+0.70889
2/3	no M , Γ	-52.39480(7574)	-52.39973(1673)	700.60	+0.32581
2/3	outer P_0	-52.71634(2739)	-52.44547(1317)	691.21	+0.28007
2/3	outer P_1	-51.57721(3175)	—	681.83	+1.14833
2/3	outer P_2	-51.04969(5593)	—	39803.45	+1.67584

Both baseline selection stages favor no M , Γ . At $\nu = 1/3$, C3-QNS validates at -36.71907(1850) meV per particle, 0.06410 meV above no M and 0.67416 meV above five-band ED. Its training value at step 116 was -37.21882 meV per particle, so the training-validation shift and large validation variance expose persistent-walker sampling lag. The precise no- M results remain above five-band ED by 0.6101 and 0.3258 meV per particle. Thus none of the present states establishes a variational energy advantage, and the strong internal C3 constraint does not improve the best validated $\nu = 1/3$ baseline. The fresh-chain outer-projector scan instead orders the sectors as $P_0 < P_2 < P_1$. Revalidating only the selected P_0 checkpoint with 4,128 new walkers gives -37.194949(10528) meV per particle and variance 81.901 meV². This is 0.41178 meV below no M and 0.47588 meV below internal C3-QNS on equal-size precise ensembles. It remains At $\nu = 2/3$, the low-cost outer scan selects P_0 : $P_0 < P_1 < P_2$ on fresh 1,024-walker chains. The apparent step-120 P_2 training advantage does not survive re-equilibration; its low-cost variance rises to 3.98×10^4 meV². Precise P_0 validation gives -52.445466(13171) meV per particle, 0.04574 meV below the precise no- M result but still 0.280073(13171) meV above five-band ED. The shift from the low-cost P_0

estimate, -52.71634(2739) meV, shows that one short selection chain is not itself a production energy estimate. This sector selection is a numerical checkpoint choice, not a claim that $m = 0$ is the unique physical $\nu = 2/3$ ground state. Part I finds an approximately threefold Γ manifold, so the three C_3 characters are expected to represent components of the same finite-size topological manifold. The present single-seed VMC has not resolved their splitting with controlled error: in particular, the anomalous P_2 validation variance precludes interpreting the raw training ordering. 0.198284(10528) meV per particle above five-band ED, so it is the best independently validated QNS here but does not establish an energy below the multiband benchmark.

19 Post-training one- and two-body observables

Figures 7 and 8 display full M and no M at both fillings, together with internal C3-QNS and outer P_0 at $\nu = 1/3$. The independently selected outer P_0 state is also shown at $\nu = 2/3$. Full- M panels use independent 1,024-walker low-cost ensembles; no- M , internal C3-QNS and selected outer P_0 use 4,128-walker precise ensembles. No persistent training configuration enters any displayed observable.

19.1 Full-ratio 1-RDM and fixed bare-band projector

For $a = (n, \mathbf{k})$ and $b = (m, \mathbf{q})$, the estimator is

$$\begin{aligned} \rho_{ab} &= \langle c_b^\dagger c_a \rangle \\ &= 2N_e \mathbb{E}_{X \sim |\Psi|^2, i, x'} \left[(\phi_a^0(x_i))^* \phi_b^0(x') \left(\frac{\Psi(X_{i \rightarrow x'})}{\Psi(X)} \right)^* \right]. \end{aligned} \quad (77)$$

The factor two accounts for uniform auxiliary sampling of the two layers. In every ratio, replacing one particle triggers a complete new ansatz call: the code rebuilds the graph and recomputes all particles' $\delta \mathbf{r}_j$, $\mathbf{J}_{\mathbf{k}j}$, the optional MQ contraction and the determinant. The projector is instead the fixed bare function

$$\phi_{n\mathbf{k}}^0(\mathbf{r}, l) = \sum_{\mathbf{G}} u_{n\mathbf{k}\mathbf{G}}^l e^{i(\mathbf{k}+\mathbf{G}) \cdot \mathbf{r}}; \quad (78)$$

no backflow phase is multiplied into it. Thus neural dressing appears once, only through the complete wavefunction ratio.

All 4,128 validation configurations and 16 auxiliary points per configuration are used to estimate the 45×45 five-band-projected matrix. The band populations are

$$W_n = \frac{1}{N_e} \sum_{\mathbf{k}} \rho_{n\mathbf{k}, n\mathbf{k}}. \quad (79)$$

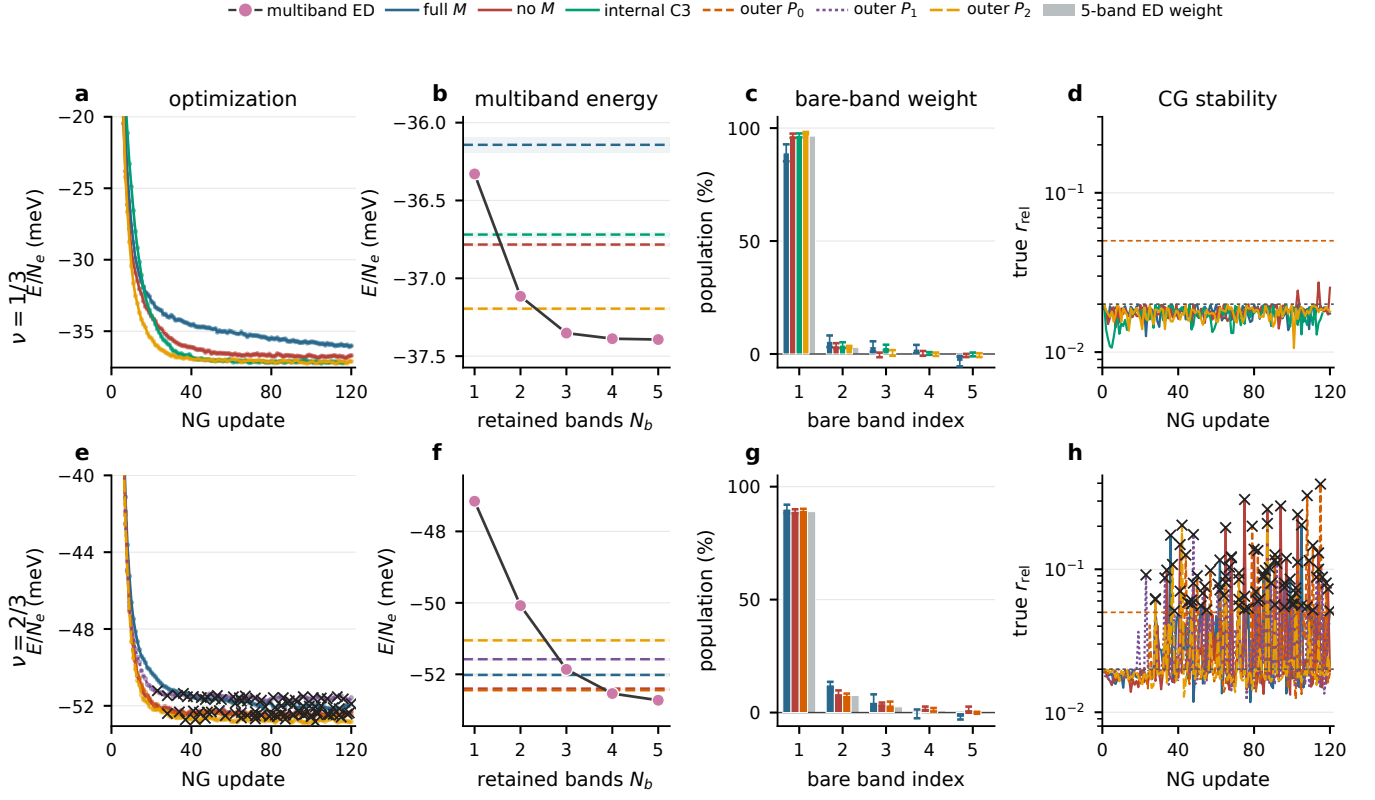


Figure 6: Controlled neural-Bloch optimization, multiband energy, band mixing and CG stability. Top row: $\nu = 1/3$; bottom row: $\nu = 2/3$. Blue, red and green denote full M , no M and internal C3-QNS. Distinct orange, purple and vermillion line styles distinguish outer P_0, P_1, P_2 . Internal C3 exists only in the upper row and stops at update 116; every outer sector reaches 120 attempted updates. (a,e) Persistent-walker training energies. (b,f) One- through five-band ED energies and independently re-equilibrated validation levels. (c,g) Bare-band populations from the complete-wavefunction-ratio 1-RDM for independently selected P_0 , with five-band ED in gray. (d,h) Explicit true CG residuals; horizontal lines mark the 0.02 early-stop and 0.05 cap-acceptance thresholds. Error bands/bars are block estimates. Training minima are not final energies.

Table 9: Five-band-projected 1-RDM results for the selected states. Band weights are percentages; errors are eight-block estimates.

ν	trace	W_1	W_2	W_3	W_4	W_5	min eig.
1/3 no M	2.986(95)	96.63(94)	3.70(1.08)	-0.40(1.02)	0.27(1.06)	-0.65(71)	-0.193
1/3 C3-QNS	3.107(67)	96.74(91)	3.94(1.26)	2.87(1.23)	0.22(60)	-0.20(84)	-0.117
1/3 outer P_0	3.021(62)	97.80(48)	3.07(51)	0.46(1.25)	-0.11(74)	-0.51(85)	-0.122
2/3 no M	6.266(176)	89.12(86)	8.38(1.42)	3.77(65)	1.87(77)	1.29(1.34)	-0.172
2/3 outer P_0	6.091(171)	89.43(76)	7.59(80)	3.40(1.44)	1.21(82)	-0.12(43)	-0.192

The projected traces are statistically compatible with N_e at the stated precision, but the matrix is only Hermitized, not projected onto the positive semidefinite cone. C3-QNS has five-band trace 3.107(67) and 3.26% nominal remote-band weight, but its minimum eigenvalue remains negative. Negative eigenvalues and the small negative $\nu = 1/3$ remote-band weights expose remaining auxiliary-coordinate noise. Consequently, no natural-orbital eigenspace is interpreted as stable. The more robust diagonal populations indicate approximately 3.4% and 10.9% nominal remote-band weight for no M at $\nu = 1/3$ and $2/3$, respectively. Outer P_0 at $\nu = 1/3$ has projected trace 3.021(62) and $W_1 = 97.80(48)\%$, corresponding to 2.20% nominal remote-band weight. Its min-

imum eigenvalue -0.122 again forbids a stable natural-orbital interpretation despite the well-normalized diagonal populations. At $\nu = 2/3$, selected outer P_0 has projected trace 6.091(171), $W_1 = 89.43(76)\%$ and 10.57% nominal remote-band weight. Its minimum eigenvalue -0.192 again shows that the current sample is insufficient for a stable natural-orbital eigenspace interpretation.

The focused bare-Slater regression sets $\delta \mathbf{r} = 0$, $J = 1$, fixes a complex 2×9 matrix M with orthonormal rows, and samples the resulting single determinant. In a complete nine-orbital test basis the exact matrix is $\rho = M^\top M^*$, with trace two. With 4,096 walkers, 100 burn sweeps and 32 auxiliary draws, the measured trace is 2.01047 ± 0.01459 , the Frobenius error is 0.05272, the Hermiticity residual is zero, and the minimum sampled eigenvalue is -0.02072 . This unit test verifies occupation, Hermiticity, PSD within sampling noise and full-basis trace normalization.

19.2 Full structure factor, connected part, and density order

The structure factor is evaluated directly as a two-body observable, not reconstructed from the 1-RDM:

$$S_{\text{full}}(\mathbf{q}) = \frac{\langle |\rho_{\mathbf{q}}|^2 \rangle}{N_e}, \quad (80)$$

$$S_c(\mathbf{q}) = \frac{\langle |\rho_{\mathbf{q}}|^2 \rangle - |\langle \rho_{\mathbf{q}} \rangle|^2}{N_e}, \quad (81)$$

$$m_{\mathbf{q}} = \frac{|\langle \rho_{\mathbf{q}} \rangle|}{N_e}. \quad (82)$$

The ED comparison uses S_{full} with the same normalization. For precise no M , the largest nonzero-momentum values are 1.5060 ± 0.0118 at $\nu = 1/3$ and 0.6760 ± 0.0106 at $\nu = 2/3$, compared with five-band ED values 1.4381 and 0.6197. C3-QNS gives $S_{\text{full}}^{\text{max}} = 1.4885 \pm 0.0095$ and $\max_{q \neq 0} m_q = 0.0108$ at $\nu = 1/3$. Outer P_0 gives $S_{\text{full}}^{\text{max}} = 1.5071 \pm 0.0089$ and $\max_{q \neq 0} m_q = 0.00795 \pm 0.00362$. The corresponding no- M maximum nonzero- q one-body amplitudes are only 0.0073 and 0.0039. At $\nu = 2/3$, outer P_0 gives $S_{\text{full}}^{\text{max}} = 0.7611 \pm 0.0131$ and $\max_{q \neq 0} m_q = 0.00550 \pm 0.00213$. Therefore the Γ states have negligible static translation-breaking expectation at the sampled reciprocal vectors; their large S_{full} peaks are density fluctuations. The real-space panels in Figs. 7 and 8 include the intra-moiré-cell Bloch density and should not be identified with a nonzero CDW order parameter by visual contrast alone.

20 Conclusion

The implemented baseline follows the printed Luo–Fu Appendix-B structure: fixed mean-field band-1 plane-wave coefficients, periodic node and edge features, trainable random auxiliary features, two unshared width-32 message iterations with the literal self-inclusive $\sum_j m_{ij}$, coordinate backflow, direct complex $J_{\mathbf{k}i}$ applied exactly once, and one final generalized Slater determinant. Full M mixes momentum sectors, whereas no M uses a fixed total- Γ orbital set.

C3-QNS adds an internally equivariant Reynolds-averaged backflow/orbital map, Bloch-sewing compensation at the dressed coordinate, and a tied sum of three C_3 -related minors. The independent character test reaches a maximum residual 7.12×10^{-5} , verifying the intended $m = 0$ symmetry. It is not an outer projection of an unrestricted determinant; it is a stronger internal inductive bias. The new outer construction instead leaves that raw network unrestricted and projects the complete generalized determinant with P_m . Character and projector-algebra residuals are below 6.4×10^{-14} and 7.6×10^{-15} , all three projected norms are nonzero, and local energy, Metropolis ratios and SR derivatives all use the projected state. It adds no trainable parameter to the 78,100-parameter no- M model.

The stopped update-116 C3 training chain reaches

−37.21882 meV per particle, but independent precise validation gives only −36.71907(1850) meV with a large local-energy variance. This is 0.06410 meV above no M and 0.67416 meV above five-band ED. The training–validation gap is direct evidence that the persistent walkers had not tracked the evolving state well enough, so continued optimization of the same chain was not justified. The outer scan selects P_0 and its independent precise energy is −37.194949(10528) meV, 0.41178 meV below no M but still 0.198284(10528) meV above five-band ED. At $\nu = 2/3$, outer P_0 validates at −52.445466(13171) meV, 0.04574 meV below no M but 0.280073(13171) meV above five-band ED. Thus outer projection improves the best independently validated energy at both fillings without yet establishing a variational energy below the multiband benchmark.

The complete-ratio 1-RDM and direct two-body structure factors remain useful diagnostics, but their non-PSD noise and the single-seed finite optimization preclude a natural-orbital or topological claim. The next controlled tests are longer, multi-seed outer- P_0 optimization, sector-selection replication, and generalized-orbital-subspace equivariance without forcing every displacement field to transform independently.

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$$\nu = 1/3$$

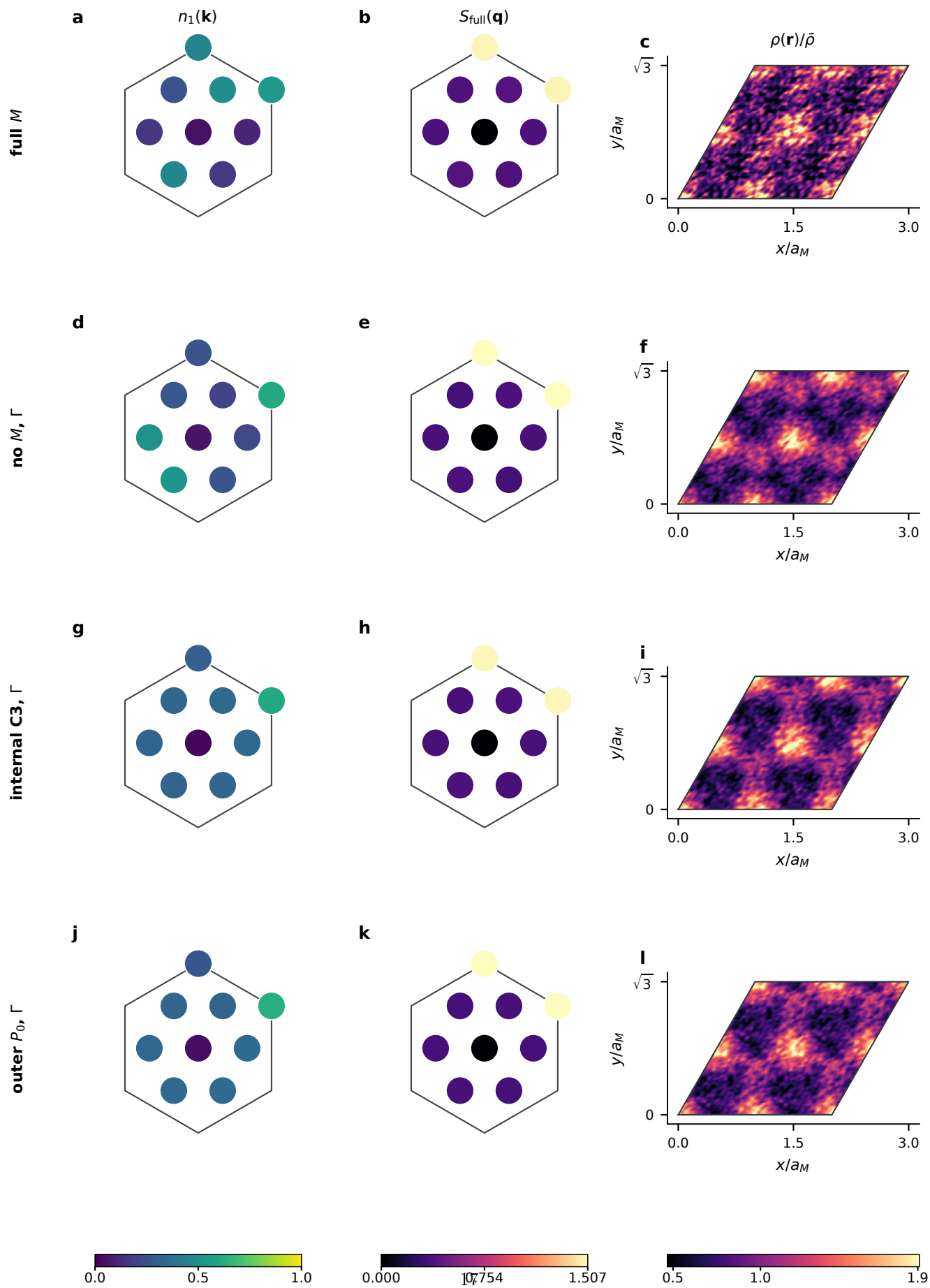


Figure 5: Effect of symmetry breaking on the density distribution. The density distribution is shown for the full M (a), no M, Γ (d), internal $C3, \Gamma$ (g), and outer P_0, Γ (j) cases. The density distribution is shown for the full M (a), no M, Γ (d), internal $C3, \Gamma$ (g), and outer P_0, Γ (j) cases. The density distribution is shown for the full M (a), no M, Γ (d), internal $C3, \Gamma$ (g), and outer P_0, Γ (j) cases.

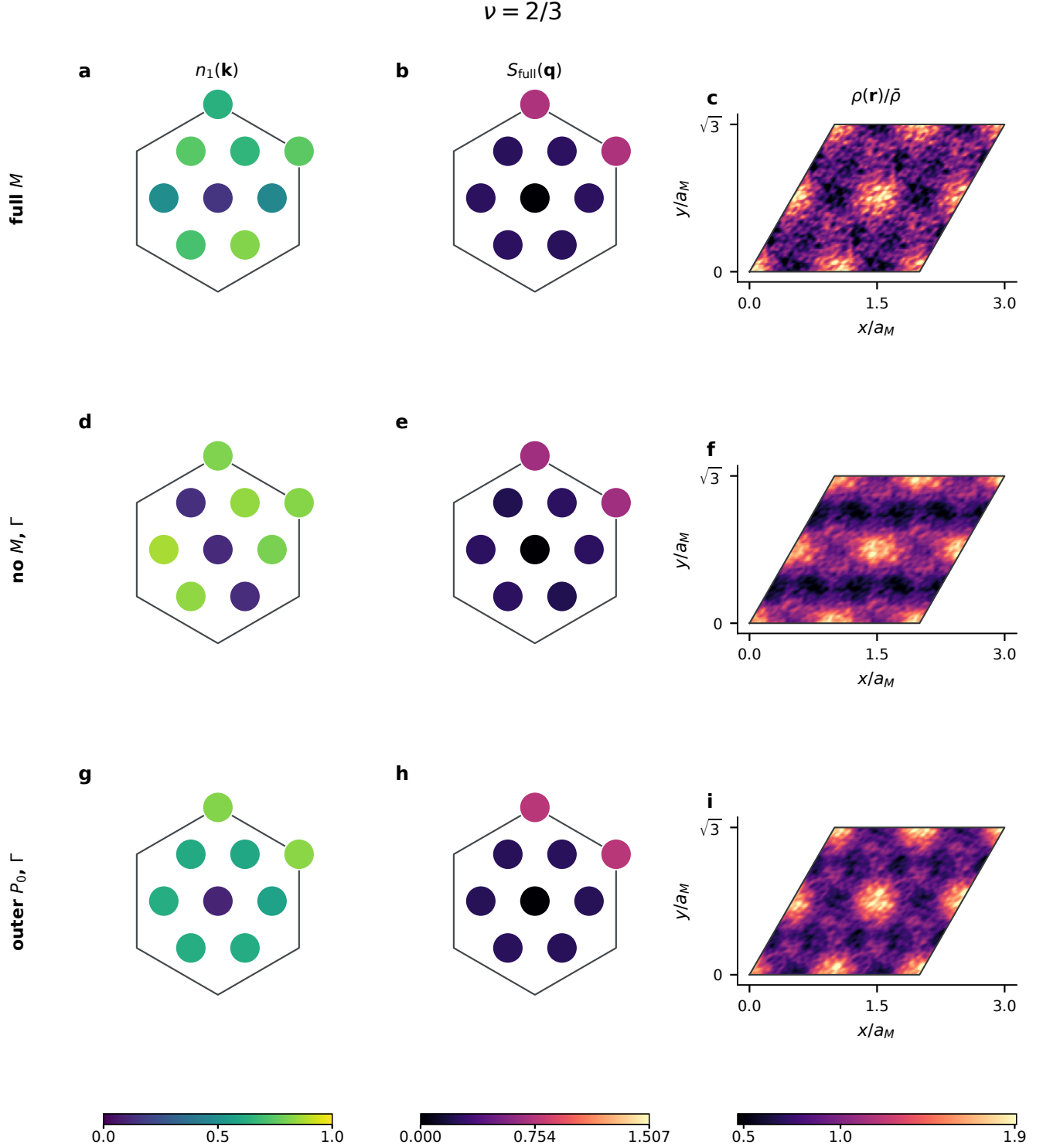


Figure 8: $\nu = 2/3$ observables from independent validation ensembles. Rows compare full M , no M and the selected outer-projector $m = 0$ state; columns use the same definitions and shared color limits as Fig. 7. The low-cost validation energies of $m = 0, 1, 2$ differ substantially, but the $m = 1, 2$ chains do not provide precise complete-ratio observable ensembles and therefore are not plotted as if their maps were statistically established. This avoids both assuming that the three sectors are identical and over-interpreting the noisy sector splitting. Full- M uses 1,024 fresh walkers; no- M and outer P_0 use precise 4,128-walker ensembles. No persistent training walker enters an estimator.